Ex: No: 1

Date: 16/7/2x.

Experiment-1.

Study of Various Network Commands in Linux and Windows:

commands:

1) arp - Address resolution protocol.
2) host name
3) ip config
4) nbt stat
5) netstat
6) ns lookup
7) path ping
8) ping - packet internet gropher
9) Route.

Command Prompt - nslookup

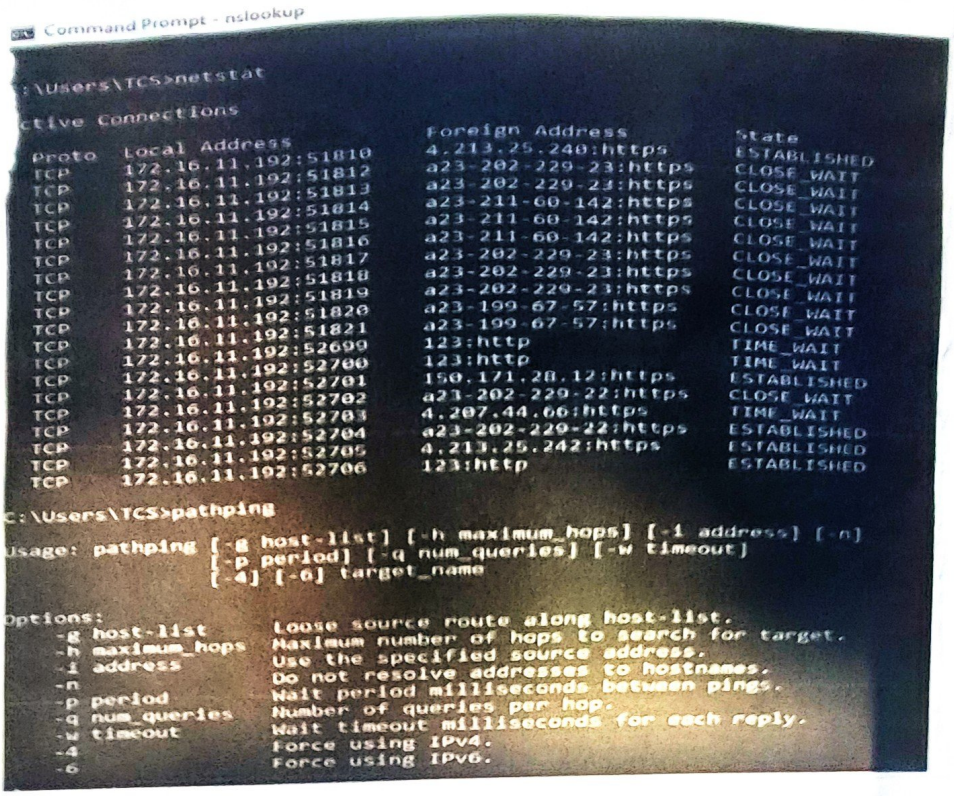
```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\TCS>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a               Displays current ARP entries by interrogating the current
                   protocol data.  If inet_addr is specified, the IP and Physical
                   addresses for only the specified computer are displayed.  If
                   more than one network interface uses ARP, entries for each ARP
                   table are displayed.
  -g               Same as -a.
  -v               Displays current ARP entries in verbose mode.  All invalid
                   entries and entries on the loop-back interface will be shown.
  inet_addr        Specifies an internet address.
  -N if_addr       Displays the ARP entries for the network interface specified
                   by if_addr.
  -d               Deletes the host specified by inet_addr. inet_addr may be
                   wildcarded with * to delete all hosts.
  -s               Adds the host and associates the Internet address inet_addr
                   with the Physical address eth_addr.  The Physical address is
                   given as 6 hexadecimal bytes separated by hyphens. The entry
                   is permanent.
  eth_addr         Specifies a physical address.
  if_addr          If present, this specifies the Internet address of the
                   interface whose address translation table should be modified.
                   If not present, the first applicable interface will be used.

Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.

C:\Users\TCS>hostname
DESKTOP-F4RM5ER
```

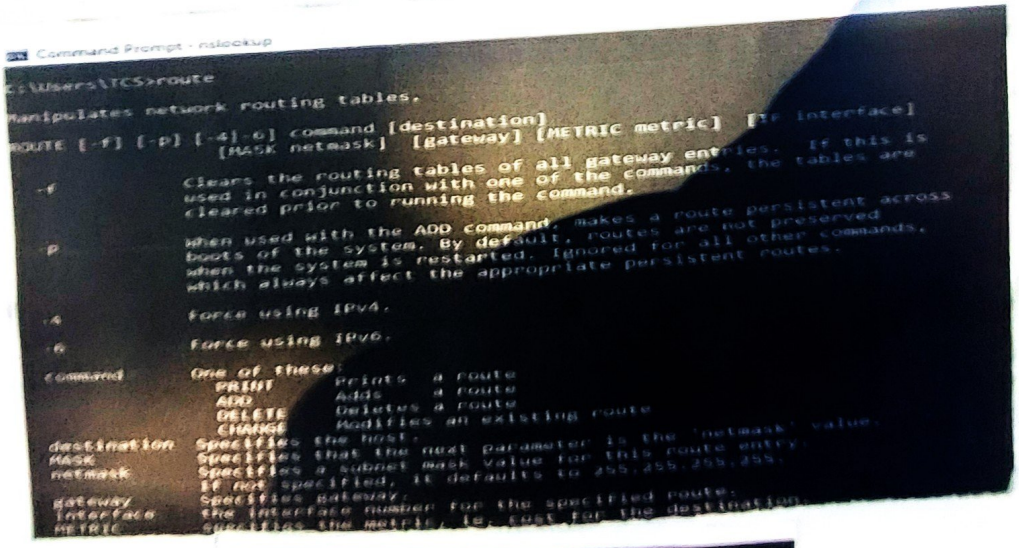
Command Prompt - nslookup

```
All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example: route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*            .... Only prints those matching 157*

    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.80.1 METRIC 3 IF 2
              destination^      ^mask      ^gateway     metric^    ^
                                                        Interface^
      If IF is not given, it tries to find the best interface for a given
      gateway.
    > route ADD 3ffe::/32 3ffe::1

    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 2

      CHANGE is used to modify gateway and/or metric only.

    > route DELETE 157.0.0.0
    > route DELETE 3ffe::/32
```

```
C:\Users\TCS>hostname
DESKTOP-F4RN5ER

C:\Users\TCS>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::d51f:7aad:f478:5629%7
   IPv4 Address. . . . . . . . . . . : 172.16.11.192
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1

C:\Users\TCS>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

```
C:\Users\TCS>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t                 Ping the specified host until stopped.
                       To see statistics and continue - type Control-Break;
                       To stop - type Control-C.
    -a                 Resolve addresses to hostnames.
    -n count           Number of echo requests to send.
    -l size            Send buffer size.
    -f                 Set Don't Fragment flag in packet (IPv4-only).
    -i TTL             Time To Live.
    -v TOS             Type Of Service (IPv4-only. This setting has been deprecated
                       and has no effect on the type of service field in the IP
                       Header).
    -r count           Record route for count hops (IPv4-only).
    -s count           Timestamp for count hops (IPv4-only).
    -j host-list       Loose source route along host-list (IPv4-only).
    -k host-list       Strict source route along host-list (IPv4-only).
    -w timeout         Timeout in milliseconds to wait for each reply.
    -R                 Use routing header to test reverse route also (IPv6-only).
                       Per RFC 5095 the use of this routing header has been
                       deprecated. Some systems may drop echo requests if
                       this header is used.
    -S srcaddr         Source address to use.
    -c compartment     Routing compartment identifier.
    -p                 Ping a Hyper-V Network Virtualization provider address.
    -4                 Force using IPv4.
    -6                 Force using IPv6.
```


Command Prompt - nslookup
:\Users\TCS>netstat
ctive Connections
Proto Local Address Foreign Address State
TCP 172.16.11.192:51810 4.213.25.240:https ESTABLISHED
TCP 172.16.11.192:51812 a23-202-229-23:https CLOSE_WAIT
TCP 172.16.11.192:51813 a23-202-229-23:https CLOSE_WAIT
TCP 172.16.11.192:51814 a23-211-60-142:https CLOSE_WAIT
TCP 172.16.11.192:51815 a23-211-60-142:https CLOSE_WAIT
TCP 172.16.11.192:51816 a23-211-60-142:https CLOSE_WAIT
TCP 172.16.11.192:51817 a23-202-229-23:https CLOSE_WAIT
TCP 172.16.11.192:51818 a23-202-229-23:https CLOSE_WAIT
TCP 172.16.11.192:51819 a23-202-229-23:https CLOSE_WAIT
TCP 172.16.11.192:51820 a23-199-67-57:https CLOSE_WAIT
TCP 172.16.11.192:51821 a23-199-67-57:https CLOSE_WAIT
TCP 172.16.11.192:52699 123:http TIME_WAIT
TCP 172.16.11.192:52700 123:http TIME_WAIT
TCP 172.16.11.192:52701 150.171.28.12:https ESTABLISHED
TCP 172.16.11.192:52702 a23-202-229-22:https CLOSE_WAIT
TCP 172.16.11.192:52703 4.207.44.66:https TIME_WAIT
TCP 172.16.11.192:52704 a23-202-229-22:https ESTABLISHED
TCP 172.16.11.192:52705 4.213.25.242:https ESTABLISHED
TCP 172.16.11.192:52706 123:http ESTABLISHED
C:\Users\TCS>pathping
Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
[-p period] [-q num_queries] [-w timeout]
[-4] [-6] target_name
Options:
-g host-list Loose source route along host-list.
-h maximum_hops Maximum number of hops to search for target.
-i address Use the specified source address.
-n Do not resolve addresses to hostnames.
-p period Wait period milliseconds between pings.
-q num_queries Number of queries per hop.
-w timeout Wait timeout milliseconds for each reply.
-4 Force using IPv4.
-6 Force using IPv6.


Command Prompt - nslookup
C:\Users\TCS>route
Manipulates network routing tables.
ROUTE [-f] [-p] [-4|-6] command [destination]
[MASK netmask] [gateway] [METRIC metric] [IF interface]
-f Clears the routing tables of all gateway entries. If this is used in conjunction with one of the commands, the tables are cleared prior to running the command.
-p When used with the ADD command, makes a route persistent across boots of the system. By default, routes are not preserved when the system is restarted. Ignored for all other commands, which always affect the appropriate persistent routes.
-4 Force using IPv4.
-6 Force using IPv6.
command One of these:
PRINT Prints a route
ADD Adds a route
DELETE Deletes a route
CHANGE Modifies an existing route
destination Specifies the host.
MASK Specifies that the next parameter is the 'netmask' value.
netmask Specifies a subnet mask value for this route entry.
If not specified, it defaults to 255.255.255.255.
gateway Specifies gateway.
interface the interface number for the specified route.
METRIC specifies the metric, ie. cost for the destination.

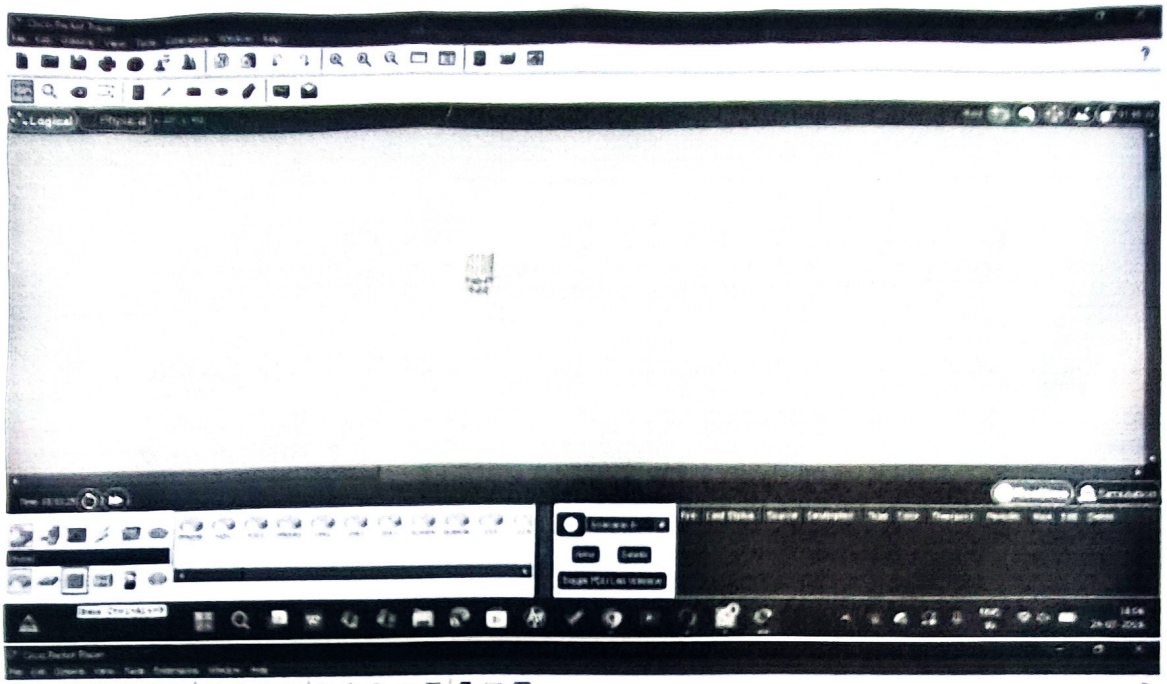
C:\Users\TCS>nslookup www.google.com
Server: UnKnown
Address: 172.16.8.1
Non-authoritative answer:
Name: www.google.com
Addresses: 2001:4860:4827:7700::
2001:4860:482b:7700::
2001:4860:4828:7700::
2001:4860:4826:7700::
2001:4860:482d:7700::
2001:4860:4829:7700::
2001:4860:482c:7700::
2001:4860:482a:7700::
142.251.156.119
142.251.153.119
142.251.157.119
142.251.152.119
142.251.151.119
142.251.154.119
142.251.155.119
142.251.150.119
C:\Users\TCS>

EX: NO: 3

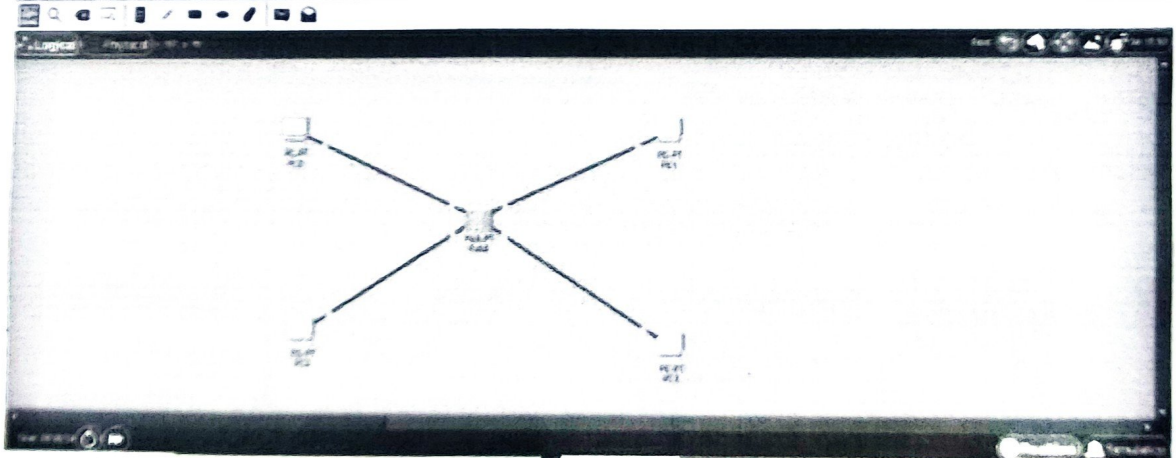
Date: 22.7.24.

Experiments on CISCO PACKET TRACKER (simulation Tool).

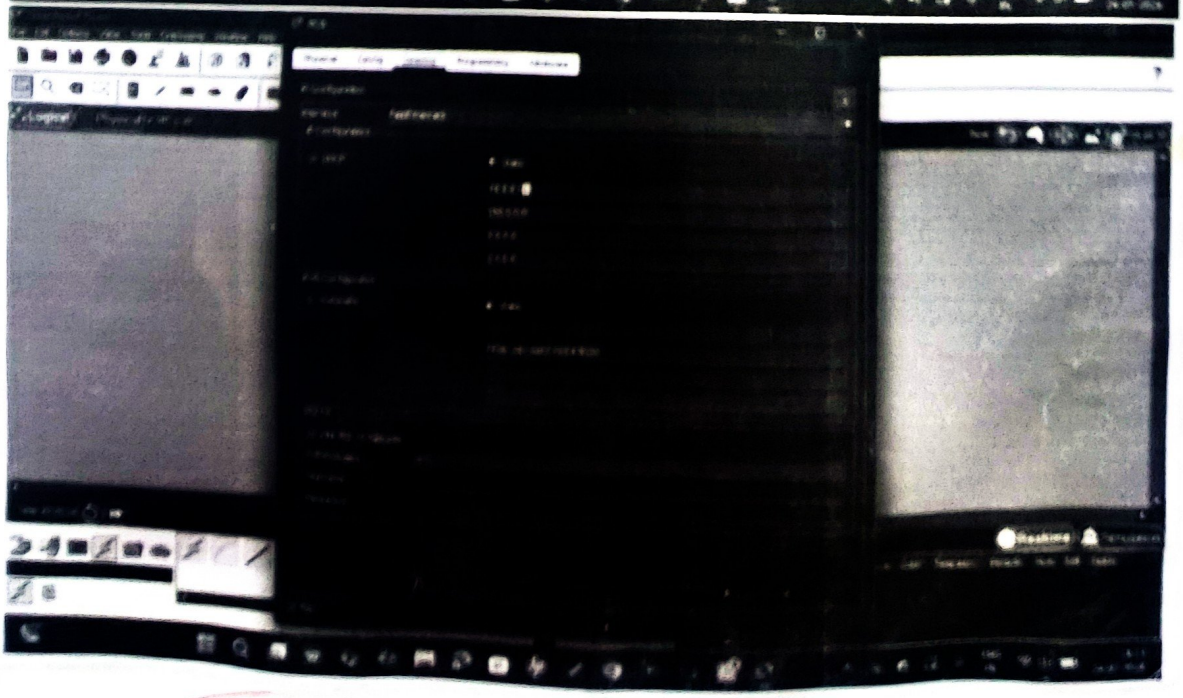
1)

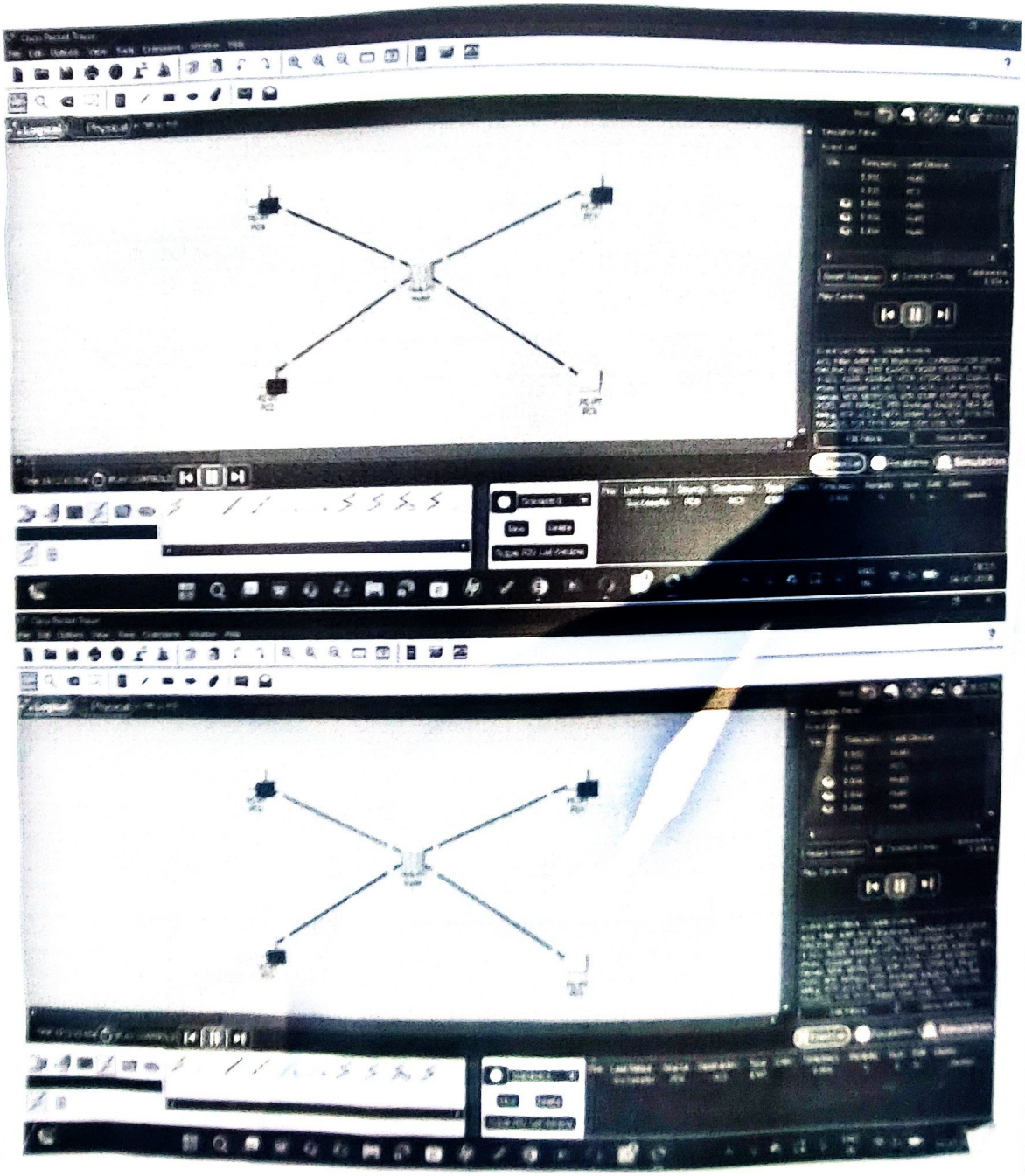


2)



3)





Result:

Procedure:

+ Downloaded & Installed Cisco Tracker 9.00
+ Opened Cisco packet tracker & used logical workspace.

* created three networks.
  - ⇒ one using HUB
  - ⇒ one using switch
  - ⇒ one using both hub & switch.
* connected 4 PCs in each network using copper straight cables.
* Assigned IP address to all PCs.
* Sent PDU from one PC to another to test communication.

HUB - sends data to all device (broadcast), causing more traffic.
Switch - sends data only to the correct device, making ~~the~~ faster and efficient.

Result

Successfully studied Cisco packet tracker, created the switch and both based network, configured IP address, and tested communication.

EXP: 2
Date:

Study of different types of network cables.

a) Understand different types of network cable.

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) cable.
2. Shielded Twisted pair (STP) cable.
3. coaxial cable.
4. Fibre Optic cable.

Cable type	category	Maximum Data Transmission	Advantages/ Disadvantages Advantages.	Application/ Use.
UTP	category 3 category 5 category 5e	10 bps Up to 100 Mbps 1 Gbps.	• cheaper in cost. • Easy to install as they have a smaller overall diameter. Disadvantages • More prone to (EMI) Electromagnetic interference and noise.	10 Base-T Ethernet Fast Ether Gigabit Ether
STP	category 6, 6a	10 Gbps	Advantages: • Shielded • Faster than UTP • Less susceptable to noise and interference. Disadvantages	Gigabit Ethernet 10G Ethern (55m) Widely u in data centres
STP	category 7	10 Gbps	• Expensive • Increases installation effort	Gigabit Ethernet, 10 Ethernet (100 m)

Coaxial Cable	RG-6 RG-59 RG-11	10-100Mbps	Advantages • High Bandwidth • Immune to interference • Low loss bandwidth • versatile Disadvantages: • Limited distance • cost • size is bulky.	Speed of signal is 500m. Television Network High Speed internet connections
fibre optics cable	Singlemode Multimode	100Gbps	Advantages • High speed • High bandwidth • High security • Long distance Disadvantages • Expensive • Requires skilled installers.	• Maximum distance fibre optics cable is around 100meters

b) Make your own Ethernet cross-over cable/straight cable:

Tools and parts needed:

• Ethernet cabling, CAT 5e is certified for Gigabit support, but CAT5 cabling works as well, just over shorter distances.

• A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.

• Two RJ45 plugs

• Optional strain relief shield

Step 1:

To start construction of the device, begin by threading shields onto the cable

Step 2:

Next, strip approximately 1.5cm of cable shielding from both ends. the crimping tool has a round area to complete this task.

Step 3:

After, you will need to untangle the wires. there should be four "twisted pairs". Referring back to the sheet, arrange them from top to bottom. One end should be in arrangement A and other in B.

Step 4:

Arrange the wires in the correct order, trim them evenly. and insert them into the RJ45 plug with the clip facing away and the gold pins, facing you.

Step 5:

Push the cable fully into the plug so the cable jacket is inside the connection. Trim and reinsert if needed.

Step 6:

Place the RJ45 plug into the crimping tool and trim it firmly without applying excessive force.

Step 7: Repeat for the other end using B for a crossover cable or A for a straight through cable. Test the cable by connecting two devices.

Result:

Thus, different types of network cables have been studied successfully.

EX: No 4. Setup and configure a LAN using a switch and Ethernet cables.

Date: 30.7.26

Aim:

To setup and configure a LAN (Local Area Network) using a switch and ethernet cables.

LAN:

A Local Area Network (LAN) refers to a network that connects devices within a limited area, such as an office building, school or home. It enables users to share resources, including data, printers and internet access. LAN connects devices to promote collaboration and to enable information between users, such as computers, printers, servers and switches. A LAN switch serves as the primary connecting device, managing and directing communication within the local network. Each connected device on a LAN switch can communicate directly with each other, allowing for fast and secure data transfer.

How to set up a LAN:

Step 1: Plan and design an appropriate network topology taking into account network requirements and equipment location.

Step 2: You can take 4 Computers, a Switch with 8 (or 24 ports which is sufficient for network of these sizes), and 4 Ethernet cables.

Step 3:

Connect your computers to network Switch via an Ethernet cable, which is as simple as plug and play one end of the Ethernet cable into your computer and the other end into your network Switch.

Step 4: Assign Ip address to your PCs:

1. Log on to the client computer as Administrator or owner.
2. Click Network and Internet connections.
3. Right click local Area Network Connection / Ethernet Go to properties -> Select Internet Protocol (TCP/IPv4) -> Click on properties -> select use the following ip address option and assign ip address.

Similarly, assign Ip address to all PCs connected to Switch: PC1 - Ip address: 10.1.1.1

PC2 - Ip address: 10.1.1.2

Step 5: Configure a Network Switch:

1. Connect your computer to the Switch: To access Switch's web Interface, you will need to connect your computer to the switch using and Ethernet cable
2. log in to the web Interface: Open a web browser and enter the IP address of the switch in address bar. This should bring up the login page for the Switch's web Interface. Enter the username and password to log in.
3. Configure basic Settings: Once you're logged in you will be able to configure basic Settings for the Switch.

4. Assign IP address as 10.1.15, subnet mask 255.0.0.0

Step6: Check the connectivity b/w switch and other machine by using ping command in the command prompt of the device.

Step 7: Select a folder → go to properties → click sharing tab → share it with everyone on the same LAN.

Result:

~~The LAN was~~ successfully set up using a switch and ~~Ethernet~~ cables IP addresses where assigned to all computers connectivity was verified using the ping command and a folder was successfully shared among accessed b/w computers on the same network.

Ex. no: 5

Date:

Packet capture using wireshark

Aim:

To capture filter inspect and analyse network packets using the wireshark packet analyzer.

Algorithm:

1. Open wireshark and launch the application
2. Select the active network interface such as wifi or Ethernet.
3. Go to capture → options and configure the capture settings
4. Set the capture limit and start the packet capture and generate network traffic
5. Observe the captured packets in the packet list
6. Apply display filters such as TCP, UDP, ARP and DNS
7. Select packets to inspect their detailed protocol information and flow graphs
8. Stop the capture 100 packets and save the capture file for future

Procedure:

* Launch wireshark and select of active network interface.
* Go to capture → options * Setup capture filter after 100 packets.
* Click start to capture packets.

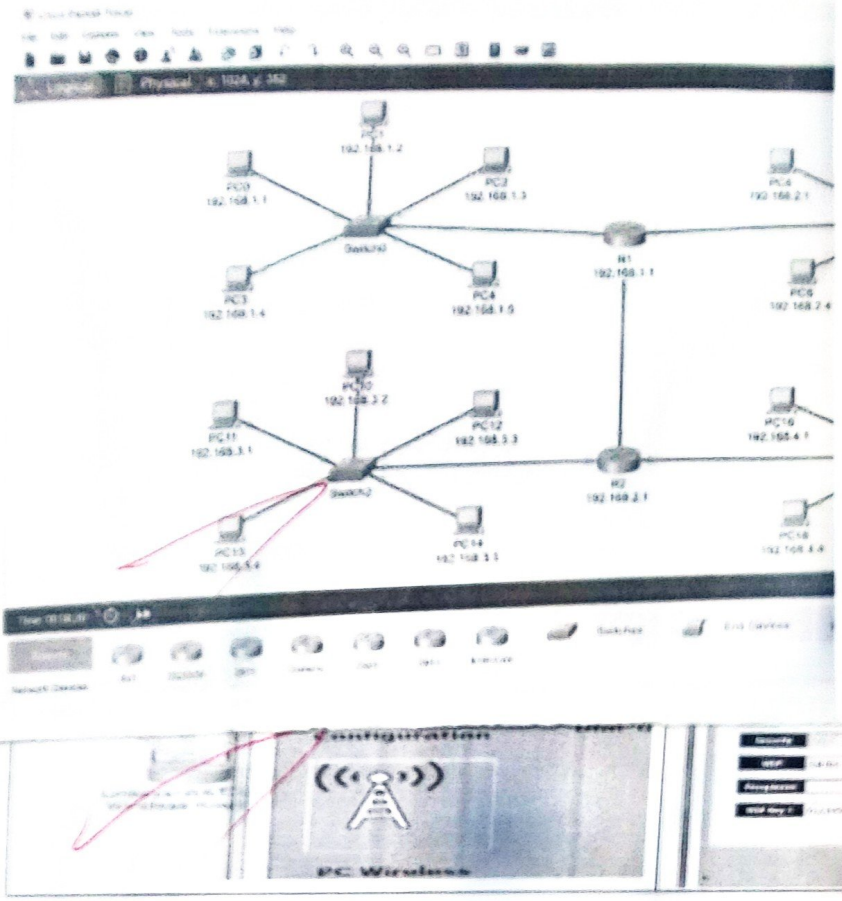
- TCP / UDP : tcp || udp

* ARP = arp
* DNS : dns
- HTTP : http

* Save the capture file to feature analysis

Result:

Successfully captured 100 network packets applied display filter to TCP, UDP analysis packet details using wireshark.



4. Calculate the number of bits required using hamming code.

5. Insert redundant bits at position 1, 2, 4, 8 ...

6. Calculate the parity bits and generate hamming code.

7. Store the generate code in the channel file.

8. At the receiver, read the data from the channel file.

9. calculate the parity bits again to find the error position.

10. If the error portion is non-zero, cannot the bit at the position.

Ex. No 6

Date:

Implementation of Error detection correction using Hamming code

Aim:

→ To write a program to implement error detection and correction using Hamming code

Concept. Make a test run to input data Stream and verify error correction feature.

Algorithm:

1. Start
2. Read the test data from the sender
3. Convert each character of the text into a bit ASCII binary form.
4. Calculate the number of redundant bits required using hamming code.
5. Insert redundant bits at position 1, 2 ...
6. Calculate the parity bits and generate hamming code
7. Store the generate code in the channel file.
8. At the receiver, read the data from the channel file.
9. calculate the parity bits again to find the error position.
10. If the error portion is non-zero, cannot the bit at the position

Ex No: 7 --

Date: Flow control at the Data link layer using Sliding window protocol

Aim:

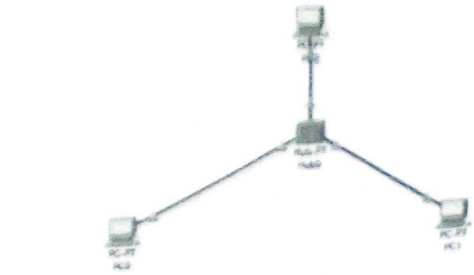
To Implement flow control at the Data link layer using the Sliding window protocol and Simulate the transmission of frames from one node to another.

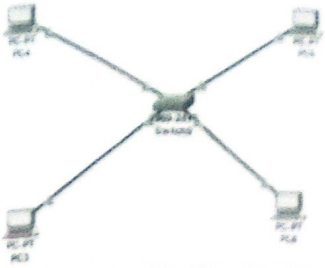
Algorithm:

1. Start the program
2. Set the total number of frames and window size
3. Send or Sends frames equal to the current window size
4. Received receives the frames and Sends acknowledgement
5. After receiving Acks, the Send or Slides the window forward
6. continue untill all frames are transmitted
7. Display the frame transmission and acknowledgement process
8. Stop.

Code:

```
def Sliding window (total_frames, window size)
    print (Sliding window protocol)
    print (
```





Network Connections

Control Panel › Network and Internet › Network Connec

Ethernet Properties

Networking Sharing

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

Obtain an IP address automatically

Use the following IP address:

IP address:

Subnet mask:

Default gateway:

Obtain DNS server address automatically

Use the following DNS server addresses:

Preferred DNS server:

Alternate DNS server:

Validate settings upon exit

Advanced...

IP Configuration

DHCP

Static

IP Address

Subnet Mask

Default Gateway

DNS Server

10.1.1.1

255.0.0.0

IP Configuration

IP Configuration

DHCP

Static

IP Address

Subnet Mask

Default Gateway

DNS Server

10.1.1.2

255.0.0.0

Capture

```
while sender < total frames:
    end = min(sender + window_size,
    print("In Sender Sends frames")
    for frame in range(sender, end):
        print("frame is frame but")
    print("Receiver sends acknowledgement:")
for frame in range(sender, end):
    print("ACK", frame, "received")

    Sender = end
    if sender < total_frames:
        print("window slides forward")
print("In All frames transmitted Successfu
Total frames = int(input("Enter total numbe
                of frames"))
window size = int(input("total frames, window
Sliding_window(total frames, window_size.
```

output:

```
Enter total number of frames = 8
Enter window size = 3
Sliding window protocol
---------
Sender Sender frames
frame 0 sent
frame 1 sent
frame 2 sent
```

Receiver sends acknowledgements:

Ack 0 received
Ack 1 received
Ack 2 received
window slides forward

Sender sends frame

frame 3 sent
frame 4 sent
frame 5 sent

Received sends Acknowledgement

Ack 3 received
Ack 4 received
Ack 5 received
window slider forward ---

Sender sends frames:

frame 6 sent
frame 7 sent

Receiver sends acknowledgements:

Ack 6 received
Ack 7 received
All frames transmitted successfully.

Result:

The sliding Window Protocol was successfully implement in python.

Send Packet

From:

computer1

To:

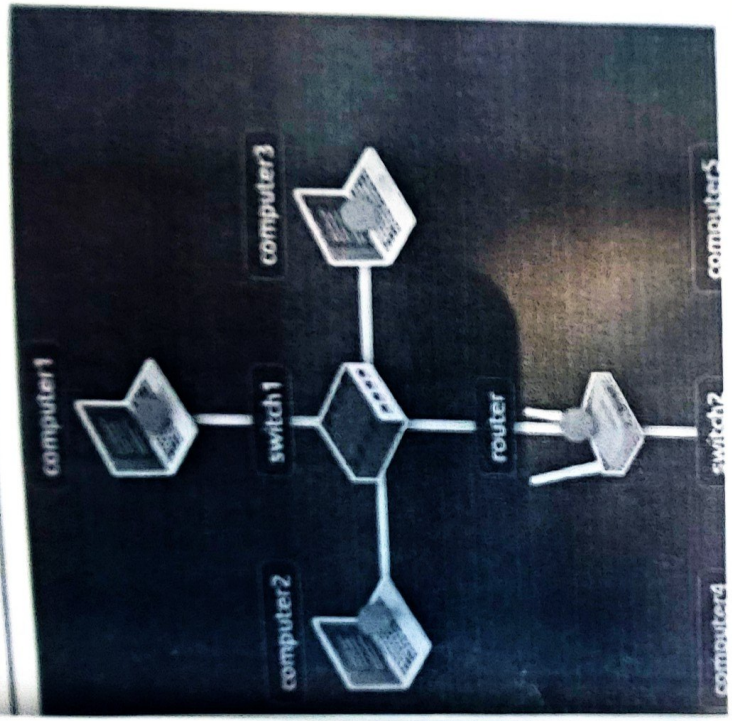
computer1

Packet Type:

arp_request

Data:

computer6



Send Packet

From:

computer4

To:

computer4

Packet Type:

arp_request

Data:

computer6

Exp: No: 8(a)

Date:

Nmap Live Host Discovery ARP, ICMP, TCP and UDP

AIM:

To discover live hosts using Nmap Scans such as ARP, ICMP, TCP and UDP on the TryHackMe platform

Algorithm:

1. Open the TryHackMe platform and access the Nmap Live Host Discovery room
2. Study the concept of Live host discovery before performing networking scanning.
3. Start with an ARP scan to identify live hosts on the local network
4. Observe the ARP requests and responses from the target hosts
5. Perform an ICMP scan to check whether target systems respond to ICMP requests
6. Perform a TCP SYN ping scan by sending TCP packets to the required ports
7. Perform a TCP ACK ping scan and observe the response from the target host
8. Perform UDP ping scan to determine whether the target system is reachable

9. Compare the results obtained from ARP, ICMP, TCP and UDP host-discovery methods.

10. Records the live hosts identified by Nmap and study the scan results.

Result:

Thus, the live hosts were successfully discovered using ARP, ICMP, TCP and UDP scanning techniques in Nmap.

8(b) Aim:

To design and configure a wireless LAN using cisco packet Tracer with three PCs connected to a linksys wireless Router

Algorithm:

1. Open cisco packet Tracer and create a new workspace.
2. place a linksys wireless Router and three PCs
3. Configure the routers with the required network Settings.
4. Set the wireless SSID as MotherNetwork.
5. Configure WEP security for the wireless network
6. open the wireless Settings of the first PC and Select MotherNetwork
7. Enter the security key and connect the PC to the router.
8. Repeat the Same configuration for the other two PCs.
9. Verify that all three PCs are connected to the wireless network.
10. Test communication between the Connected PCs

Result:

Thus, the wireless LAN was successfully configured using cisco packet Tracer with three PCs connected to the Linksys wireless Router.

9.

Aim:

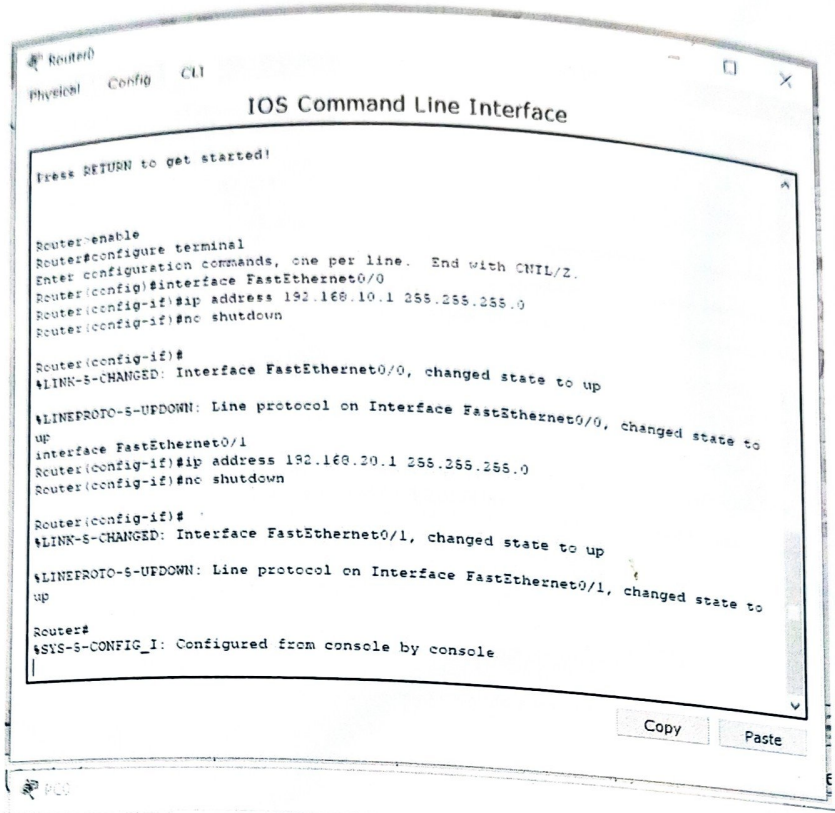
To design and configure a subnetted network using cisco packet Tracer.

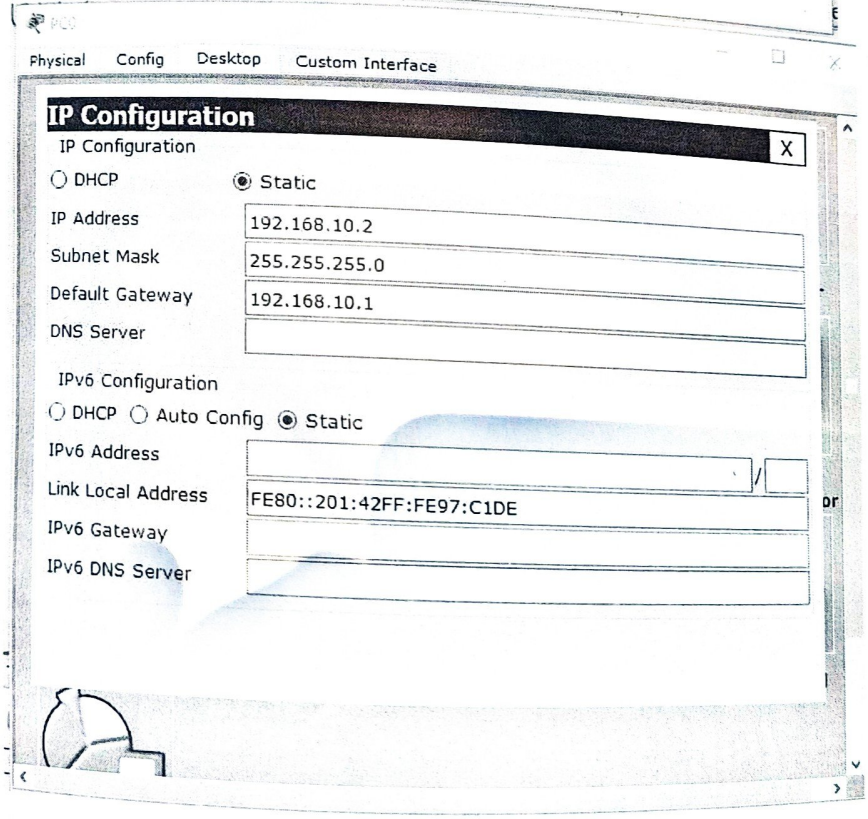
Algorithim.

1. open cisco packet Tracer.
2. Add a router, Switch and PCs
3. connect the devices using suitable cables.
4. open the router CLI
5. configure the router interfaces with IP addresses and Subnet masks
6. Enable the interface using no shutdown
7. Configure the Ip address and Subnet-mask of each pc
8. Set the appropriate default gateway for each pc
9. use the ping command to test connectivity
10. verify communication between the different subnets.

Result:

Thus, the subnetted network was Successfully configured and connectivity was verified using ping.


Router0
Physical
Config
CLI
IOS Command Line Interface
Press RETURN to get started!
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
interface FastEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste


PC0
Physical
Config
Desktop
Custom Interface
IP Configuration
IP Configuration
X
DHCP
Static
IP Address
192.168.10.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::201:42FF:FE97:C1DE
IPv6 Gateway
IPv6 DNS Server

Aim:

To design and configue an internetwork using a router in Cisco packet Tracer.

Algorithim

1. open cisco packet Tracer.
2. Add one router and two PCs
3. Connect the PCs to the router.
4. configue the router interfaces with IP addresses.
5. Enable the interfaces using no shutdown.
6. configue Ipaddress and Subnet mask for PCs
7. configue Ipaddress and Subnet- mask for PC1
8. Set the default gateways for both PCs
9. Send a PDU from PC0 to PC1
10. Verify the communication between the two networks.

Result:

Thus, the internetwork was Successfuley onfigured using a router and Connectivity s verified.

(6Cb)

Aim:

To design and configure an internetwork using wireless router, DHCP Server and internet cloud.

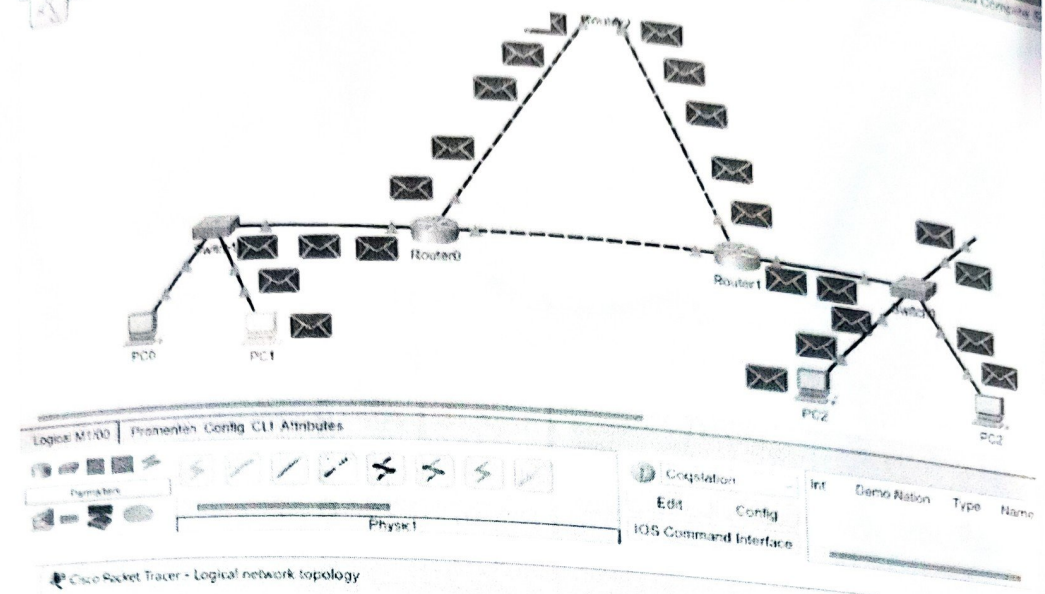
Algorithim

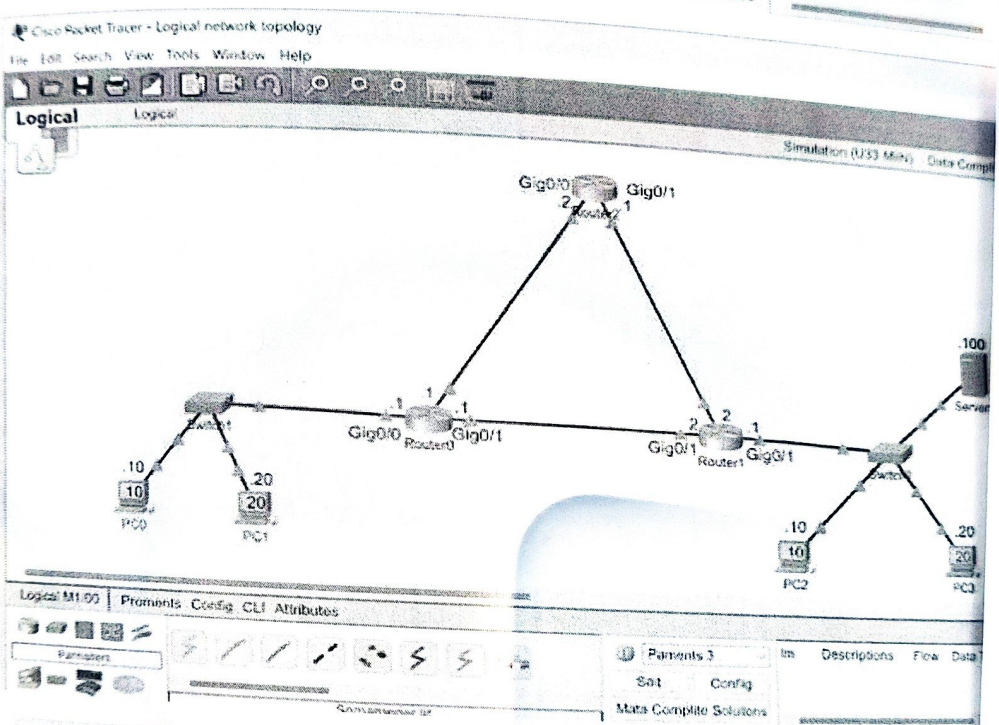
1. Open Cisco packet Tracer.
2. Add a wireless router, DHCP Server and internet cloud.
3. Add the required wireless devices.
4. Connect the device to the wireless router.
5. Configure the DHCP Server with Ip settings.
6. Configure DHCP Server with Ip Settings.
7. Enable DHCP for automatic Ip assignment
8. Connect the wireless device to the network
9. Verify the assigned Ip addresses.
10. Test connectivity between the device and internet cloud.

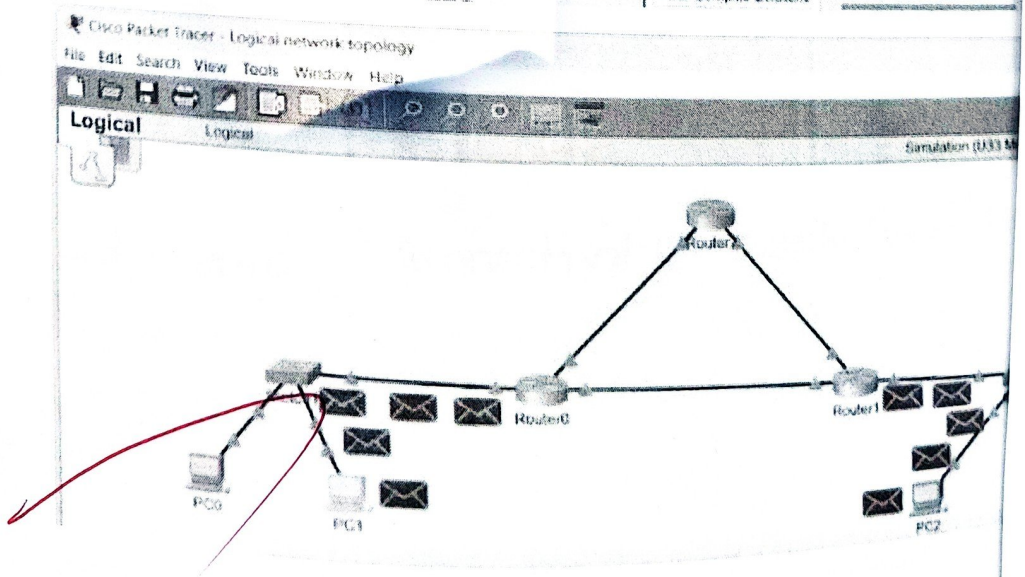
Result:

Thus, the wireless internetwork was Successfully configured and connectivity was verified.

We also configured a separate static host route for the host 30.0.0.100/8...







Aim:

To configure static, routing using cisco packet Tracer.

Algorithm.

1. open cisco packet Traces.
2. Create the required routers and pc to polog
3. Connect the devices using suitable cables.
4. Assign Ip addresses to pcs and routes interfaces
5. Enable routes interfaces using to Shutdown
6. Configure static routes using Ip route
7. Configure the required backup route
8. Check the routing table using show Ip route.
9. Test connectivity using ping
10. Verify communication between the networks

Result:

Thus, Static Outing was successfully Configured and connectivity was verified.

a(b)

Aim:

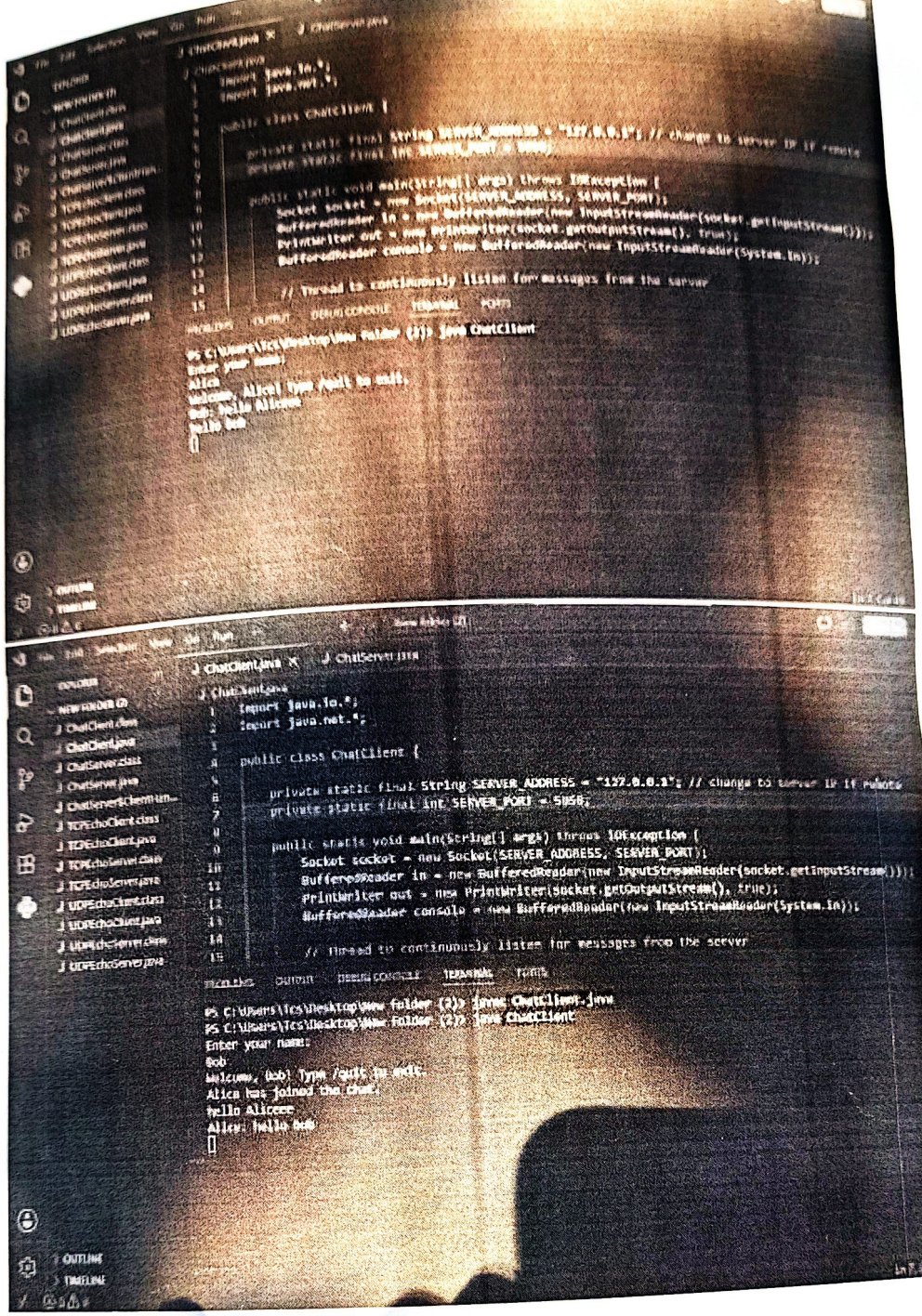
To Configure RIP routing using Cisco Packet Tracer.

Algorithm

1. Open Cisco packet Tracer.
2. Create the required routers and pc topology
3. Connect the device using suitable cables
4. Assign IP addresses to PCs and routers interfaces.
5. Enable router interfaces using to studd
6. Enter RIP configuration mode using routers IP
7. Add the connected networks using network commands.
8. Configure RIP on the required routers
9. Check the routing table
10. Test Connectivity using ping.

Result:

Thus, RIP routing was Successfully Configured and Connectivity between the network was verified.



Aim:

To implement an echo client server using TCP / UDP Sockets.

Algorithms.

1. Create the Server Socket.
2. Bind the Socket to a Ip addressed and port.
3. create the client to the Server.
4. Create the client Socket.
5. Connect the client to the Server.
6. Enter a message from the client.
7. Send the message to the Server
8. Server received and echoes the message
9. client received to echoed message
10. Close the Sockets.

Result:

Thus, the echo client - server was successfully implemented using Sockets.

2(b)

Aim:

To implement a chat program using Socket programming.

Algorithms:

1. Create the Server Socket
2. Bind the Socket to an Ip address and port
3. Start listing for a client
4. Create the client Socket
5. Establish connection between client and server.
6. Enter a message from the client
7. Send and receive message between client and server.
8. Display the received messages
9. Continue communication until termination
10. Close the sockets.

Result:

Thus the chat client - Server was Successfully implemented

```
sequence = 1
header = struct.pack(
        = !BBHHH",
          8,
          0,
          0, packet_id, sequence)
data = b"PING"
chksum = checksum(header + data)
header = struct.pack
        "!BBHHH",
        8,
        0,
        chksum,
        packet_id,
        sequence,
    )
Packet = header + data.
Start = time.time()
sock.sendto(packet, dest_addr, 1))
sock.settimeout(2)
sock.recvfrom(1024)
end = time.time()
rtt = (end - start) * 1000
Print("Reply from", dest_addr,
        ": time = %.2fms" %rtt
sock.close()
except socket.timeout:
    Print("Request timed out")
except PermissionError:
    Print("permission denied. Run the
Program as administrator (root.)
```

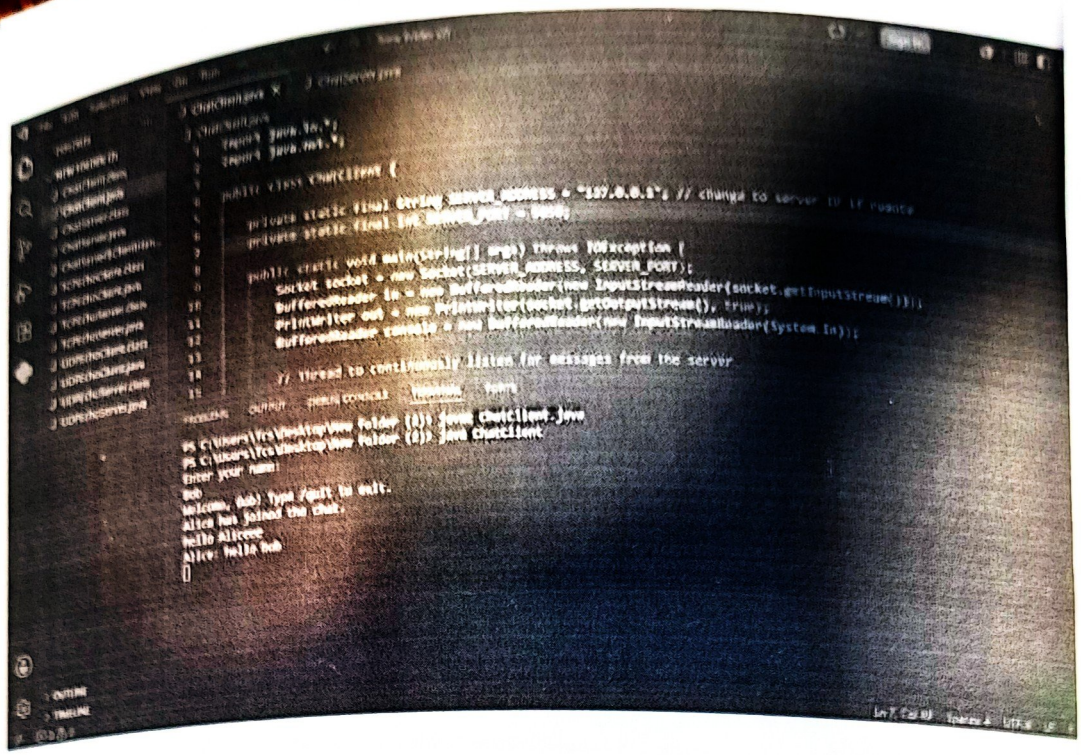
EXP NO: 13
DATE:

IMPLEMENTED YOUR OWN PING PROGRAM.

AIM:
To implement a ping program using Python raw sockets to check the reachability of a host and calculate the round trip time.

CODE:

```
import socket
import struct
import time
import os
def checksum(data):
    total = 0
    for i in range(0, len(data)-1, 2):
        total += (data[i] << 8) + data[i+1]
    if len(data) % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
    total += total >> 16
    return (~total) & 0xffff
def ping(host):
    try:
        dest_addr = socket.gethostbyname(host)
    sock = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_ICMP)
    packet_id = os.getpid() & 0xffff
```


Welcome, Bob! Type /quit to exit.
Alice has joined the chat.
Hello Alice!
Alice: hello Bob

UDPEchoClient.java
import java.io.*;
import java.net.*;
public class ChatClient {
private static final String SERVER_ADDRESS = "127.0.0.1";
private static final int SERVER_PORT = 5051;
PS C:\Users\fcs\Desktop\New folder (2)> java UDPEchoClient
Input: UDP packet test.
UDP Echo: UDP packet test.
PS C:\Users\fcs\Desktop\New folder (2)>

```
print ("-----------")
print ("Source IP :", source_ip)
print ("Destination IP :", destination_ip)
print ("Protocol :", protocol)
print ("Packet size:", len(packet), "bytes")
print ("-----------")
```

Sample Output:

```
Packet Sniffer started....
Waiting for packets....
-----------
Source IP: 192.168.1.10
Destination IP: 142.250.183.14
Protocol : 6
Packet size: 60 bytes.
-----------
Source IP: 8.8.8.8
Destination IP: 172.168.1.10
Protocol : 1
Packet size: 84 bytes.
-----------
```

Result:

Thus, the ping program and packet Sniffing Program were sucessfully implemented using python raw sockets.

EXP NO: 14
DATE:

PACKET SNIFFING USING RAW SOCKETS.

AIM:

To implement a packet Sniffing using Python raw Sockets and display the Source IP, destination IP and protocol of captured packets.

CODE:

```
import socket
import struct
sock = socket.socket(
        socket.AF_INET,
        socket.SOCK_RAW,
        socket.IPPROTO_IP )
Print ("Packet sniffer started...")
Print ("Waiting for packets...\n")
while True:
    Packet, address = sock.recvfrom(65535)
    ip_header = packet[0:20]
    iph = struct.unpack(
            "!BBHHHBBH4s4s",
            ip_header.
        )
    Source_ip = socket.inet_ntoa(iph[8])
    destination_ip = socket.inet_ntoa(iph[9])
    protocol = iph[6]
```

```
host = input ("Enter destination host:")
ping (host).
```

Sample INPUT/OUTPUT:

Enter destination host: 8.8.8.8
Reply from 8.8.8.8 : time = 24.53ms.

Result:

Thus, the ping progeam was sucessfully implemented in Python using raw Sockets.

Exp No: 15 ANALYSIS OF VARIOUS TYPE
DATE: OF SERVES USING
WEBALIZER.

AIM:

To analyze the web server log files using the Webalizer tool and study various server statistics such as number of requests, visitors, transferred data, URLs and response status codes.

ALGORITHM:

1. Start the web alizer application
2. Install and configure Webalizer on the system.
3. Locate the web server access log file
4. Configure Webalizer with the required log-file path and output directory.
5. Run Webalizer on the selected log file.
6. Analyze the server activity using the generated graphs and tables.
7. Repeat the analysis for different server log files if required.

INPUT:

Web server access log files containing information such as:

client IP.
Date and time
requested URL
HTTP Method
HTTP status code
Bytes Transfered
Referer.

Example Input:

access.log.

Output:

Webalizer generates an HTML-based statistical report contains.

Webalizer statistics.

Total request : 1250
Total Visits : 480
Total pages : 920
Total Bytes : 15.4 mb
Successful Requests : 1100
404 Bytes Error : 45

Result:

Thus the webalizer tool was used successfully to analyze web server log files, and various server statistics such as requests, visitors, pages, bandwidth usage and HTTP.